

ORIGINAL RESEARCH

Enhanced Detection and Prompt Diagnosis of Atrial Fibrillation Using Apple Watch



A Randomized Controlled Trial

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ABSTRACT

BACKGROUND Atrial fibrillation (AF), the most common cardiac arrhythmia, is a major cause of stroke and often remains undiagnosed due to its paroxysmal and frequently asymptomatic nature. Wearables provide a scalable, noninvasive screening tool.

OBJECTIVES This trial evaluated new onset AF detection in patients at elevated stroke risk using remote smartwatch-based screening.

METHODS This prospective multicenter randomized controlled trial included patients ≥ 65 years with elevated stroke risk ($\text{CHA}_2\text{DS}_2\text{-VASc} \geq 2$ men, ≥ 3 women) from 2 secondary care centers in the Netherlands. Patients were randomized to 6-month (180-day) monitoring with a smartwatch with photoplethysmography and single-lead electrocardiogram (ECG) functions, or standard care. ECGs were reviewed remotely by an independent eHealth team within 24 hours. The primary outcome was new onset AF, defined as a confirmed episode lasting ≥ 30 seconds on single-lead ECG or standard ECG methods.

RESULTS Between November 2022 and December 2023, 437 patients were randomized (219 intervention, 218 control); the median age was 75 years, 46.7% were female and the median $\text{CHA}_2\text{DS}_2\text{-VASc}$ score was 3.0. New onset AF occurred in 21 (9.6%) patients of the intervention group and 5 (2.3%) patients of the control group (risk difference: 7.3 percentage points; 95% CI: 2.9-11.7 percentage points; $P = 0.001$; HR: 4.40; 95% CI: 1.66-11.66). Several asymptomatic AF episodes were detected only in the intervention group, while paroxysmal AF occurred in both groups.

CONCLUSIONS This randomized controlled trial provides evidence that 6-month smartwatch-based AF screening enhances the detection rate of new onset AF compared with standard care in patients at elevated stroke risk. (Detection and Quantification of Atrial Fibrillation in High-risk Patients Using a Smartwatch Wearable [Apple Watch] [EQUAL]; [NCT05686330](https://doi.org/10.1016/j.jacc.2025.11.032)) (JACC. 2026;87:1714-1728) © 2026 by the American College of Cardiology Foundation.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](https://doi.org/10.1016/j.jacc.2025.11.032).

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Atrial fibrillation (AF), the most prevalent sustained cardiac arrhythmia and a major cause of stroke and heart failure worldwide.¹⁻³ Approximately one-half of AF cases are paroxysmal and often asymptomatic, leading many to remain undetected in routine clinical care.^{4,5} Current European guidelines recommend rhythm assessment in individuals at increased stroke risk, and expert consensus supports screen-detected AF in individuals aged ≥ 65 years should prompt clinical evaluation and consideration of guideline-directed management.⁶⁻⁸ Despite these recommendations, optimal approaches for large-scale, noninvasive, prolonged AF detection remain uncertain.

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Prolonged monitoring with implantable loop recorders (ILRs) can identify AF in up to 20% of high-risk patients within 6 months,⁹ but these devices are invasive, costly, and unsuitable for large-scale use.¹⁰ Studies using noninvasive screening approaches, such as handheld single-lead ECG devices allowing for intermittent rhythm assessment, have demonstrated increased AF detection compared with routine care.^{11,12} However, such devices rely on active participation and cannot continuously monitor for irregular rhythms between recordings, potentially underdiagnosing asymptomatic AF patients. Population-based screening using adhesive ECG patches provided continuous rhythm monitoring, but at the cost of limited wear time (<14 days) and potential skin irritation.^{13,14}

Wearable devices equipped with photoplethysmography (PPG) sensors provide a noninvasive method for rhythm monitoring by detecting blood volume changes through light absorption.¹⁵ Their passive nature makes them particularly suitable for prolonged monitoring, including detection of paroxysmal and asymptomatic episodes of AF. However, confirmation of AF requires physician review of an ECG, whether 12-lead or single-lead, as recommended in the 2024 European Society of Cardiology guidelines on AF management.⁷ Large siteless studies established the feasibility of PPG-based digital screening in broad, self-enrolled populations, yet confirmation of AF relied on delayed ECG verification using separate devices, thereby limiting timely diagnosis, prompt therapy initiation and integration into clinical workflows.¹⁶⁻¹⁸ Early economic modeling studies suggest that wearable AF screening in high-risk populations may be cost-effective, particularly when paired with timely ECG confirmation and clinical evaluation.^{19,20}

To date, no randomized controlled trial has evaluated smartwatch-based AF screening in a high-risk

secondary-care population using a device that integrates both PPG and ECG functionality. The EQUAL (Enhanced detection and Quickened diagnosis of Atrial fibrillation using AppLe Watch) trial was designed to address this gap by assessing the feasibility and diagnostic yield of smartwatch-based screening compared with standard care in cardiology outpatients at increased stroke risk. In addition, this trial provides the first evidence that combined PPG- and ECG-based smartwatch monitoring can be effectively implemented within a 24/7 telemonitoring infrastructure in routine specialty care. The overall study design and workflow are summarized in the [Central Illustration](#).

METHODS

STUDY DESIGN. The EQUAL trial was a multicenter, 2-arm, randomized controlled trial designed to evaluate smartwatch-based detection of AF. Patient recruitment took place at 3 sites in the Netherlands: 2 separate locations of Cardiology Centers of the Netherlands in Amsterdam, and Spaarne Gasthuis in Haarlem.

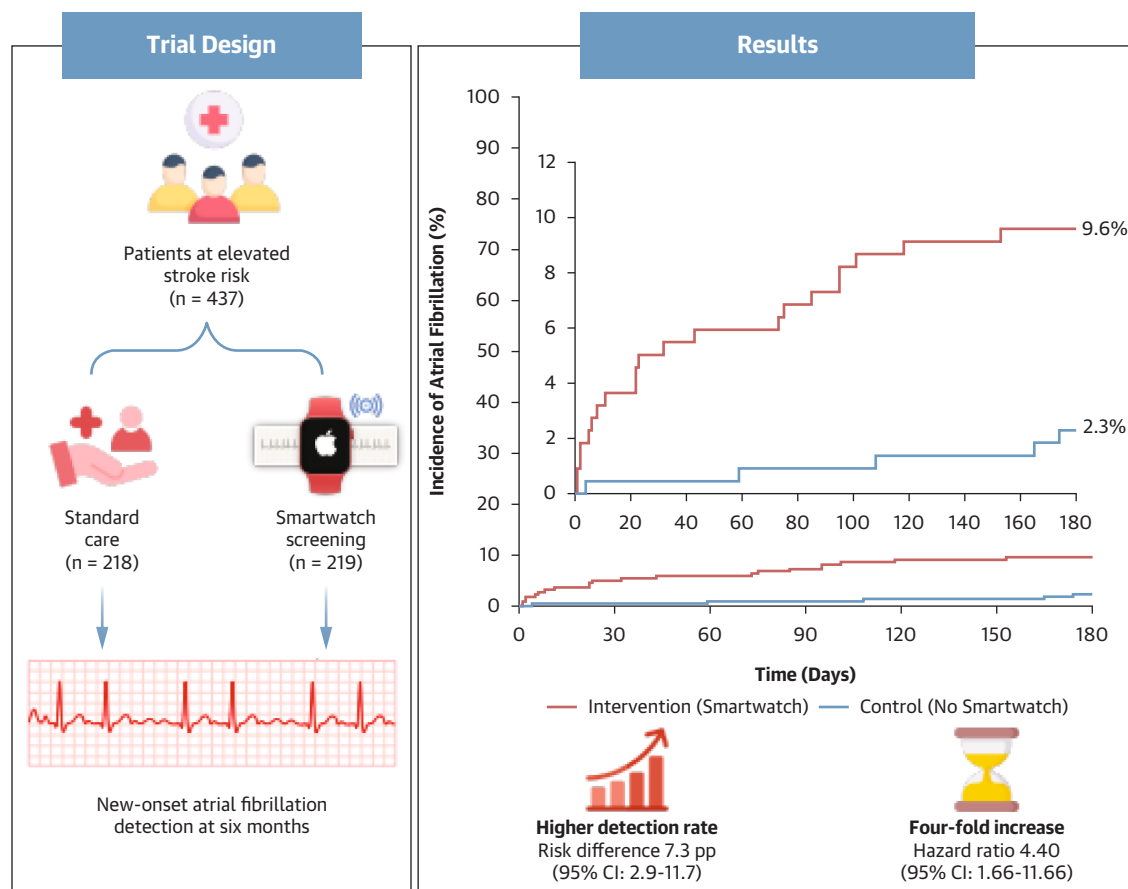
The protocol was reviewed and approved by the local ethics committee at the Amsterdam University Medical Center (reference number: NL79151.018.21) and adhered to the principles outlined in the Declaration of Helsinki. The trial was preregistered at ClinicalTrials.gov ([NCT05686330](#); Protocol ID: 2021.209). All patients provided written informed consent.

PATIENTS. Consecutive patients ≥ 65 years of age who visited the outpatient cardiology clinics at either secondary care clinic were assessed for eligibility. Inclusion required a CHA₂DS₂-VASc score of ≥ 2 for men or ≥ 3 for women, proficiency in Dutch or English, and the ability to record a smartwatch ECG after brief instruction. The CHA₂DS₂-VASc score estimates stroke risk in AF using age, sex, and comorbidity. A score of ≥ 2 for men and ≥ 3 for women indicates high stroke risk and mandates prophylactic anticoagulation in case of diagnosed AF.^{7,21} Exclusion criteria were prior history of AF or atrial flutter, cardiac implantable electronic devices (ie, pacemaker, implantable cardioverter-defibrillator, or ILR), ongoing anticoagulation, or any condition precluding smartwatch use (ie, allergic reactions, amputations, and wounds).

RANDOMIZATION. Randomization was performed 1:1 via a centralized, Web-based Castor EDC module using a computer-generated sequence with randomly permuted block sizes.^{4,6,8} Allocation concealment was ensured because the sequence and block sizes

ABBREVIATIONS AND ACRONYMS

AF	= atrial fibrillation
AW	= Apple Watch
ECG	= electrocardiogram
ED	= emergency department
ILR	= implantable loop recorder
MACE	= major adverse cardiovascular events
PPG	= photoplethysmography
PPV	= positive predictive value

CENTRAL ILLUSTRATION A Randomized Controlled Trial Investigating Smartwatch-Based Screening for Atrial Fibrillation

van Steijn NJ, et al. JACC. 2026;87(14):1714–1728.

The EQUAL (Enhanced detection and Quicken diagnosis of Atrial fibrillation using AppLe Watch) trial evaluated whether 6-month rhythm monitoring with a smartwatch improved the detection of atrial fibrillation compared with standard care. The illustration shows the randomized design, intervention pathway with smartwatch monitoring and telemedicine, and the observed increase in atrial fibrillation detection in the intervention group. Created in BioRender. pp = percentage points.

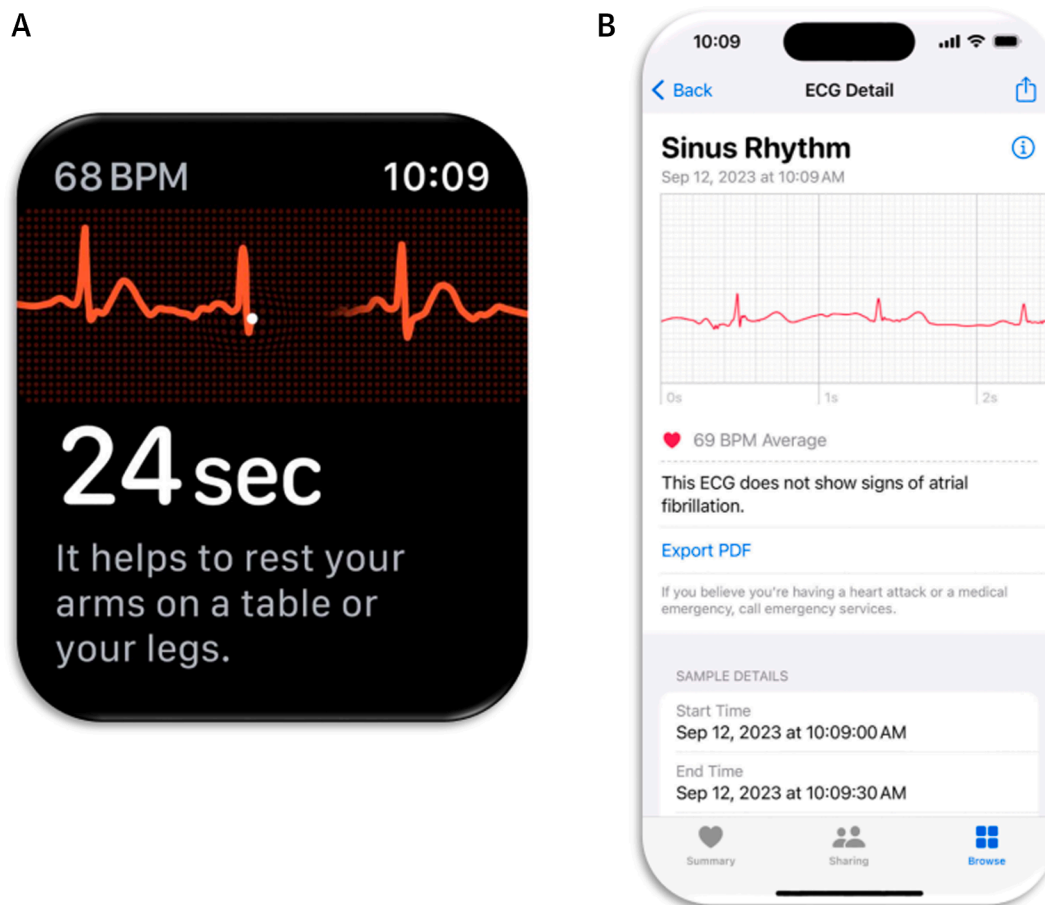
were embedded in Castor and not accessible to investigators; the assignment was released only after eligibility confirmation and informed consent. The trial was open label, with independent adjudicators.

PROCEDURES. Follow-up was predefined as 6 months (180 days) or until AF was diagnosed, whichever occurred first. Patients in the intervention group received an Apple Watch (AW) Series 5 or 8 (Apple), paired with their smartphone (Figure 1). After standardized instruction, patients were asked to wear the watch ≥ 12 hours/day while awake and to record a 30-second single-lead ECG in response to symptoms or after an irregular rhythm notification, automatically triggered when the watch detects an

irregular pulse pattern suggestive of AF. AF was diagnosed if AF was present throughout a complete 30-second recording.

All smartwatch ECGs were automatically transmitted to the HartWacht telemonitoring platform and reviewed within 24 hours by at least 1 dedicated adjudicator, a trained cardiac nurse, and supervised by a cardiologist. Adjudication followed a predefined protocol, endorsed by the Dutch Society of Cardiology and was based solely on ECG tracings, blinded to irregular rhythm notifications. Technical and procedural details are provided in the [Supplemental Methods](#). Figure 2 shows a schematic overview of the overall trial workflow, and Figure 3 illustrates the

FIGURE 1 Apple Watch ECG Recording Interface



(A) Apple Watch display during a single-lead electrocardiogram (ECG) recording, including the countdown timer. (B) The Apple Health app display after completion of the recording. A 30-second lead I-equivalent tracing can be exported as a PDF file through the app.

clinical review pathway within the HartWacht tele-monitoring platform for ECGs classified by the eHealth team as AF or atrial flutter.

Patients in the control group received standard care at clinician discretion, which could include 12-lead ECG, Holter monitoring, in-hospital telemetry, or handheld mobile ECG devices. At the end of follow-up, control patients were contacted by telephone to inquire about AF diagnoses and any rhythm monitoring performed during follow-up, including patient-initiated mobile ECG use. Patients in both groups were instructed to notify the trial team if AF was diagnosed during follow-up. In all cases, treatment decisions were made by the treating cardiologist.

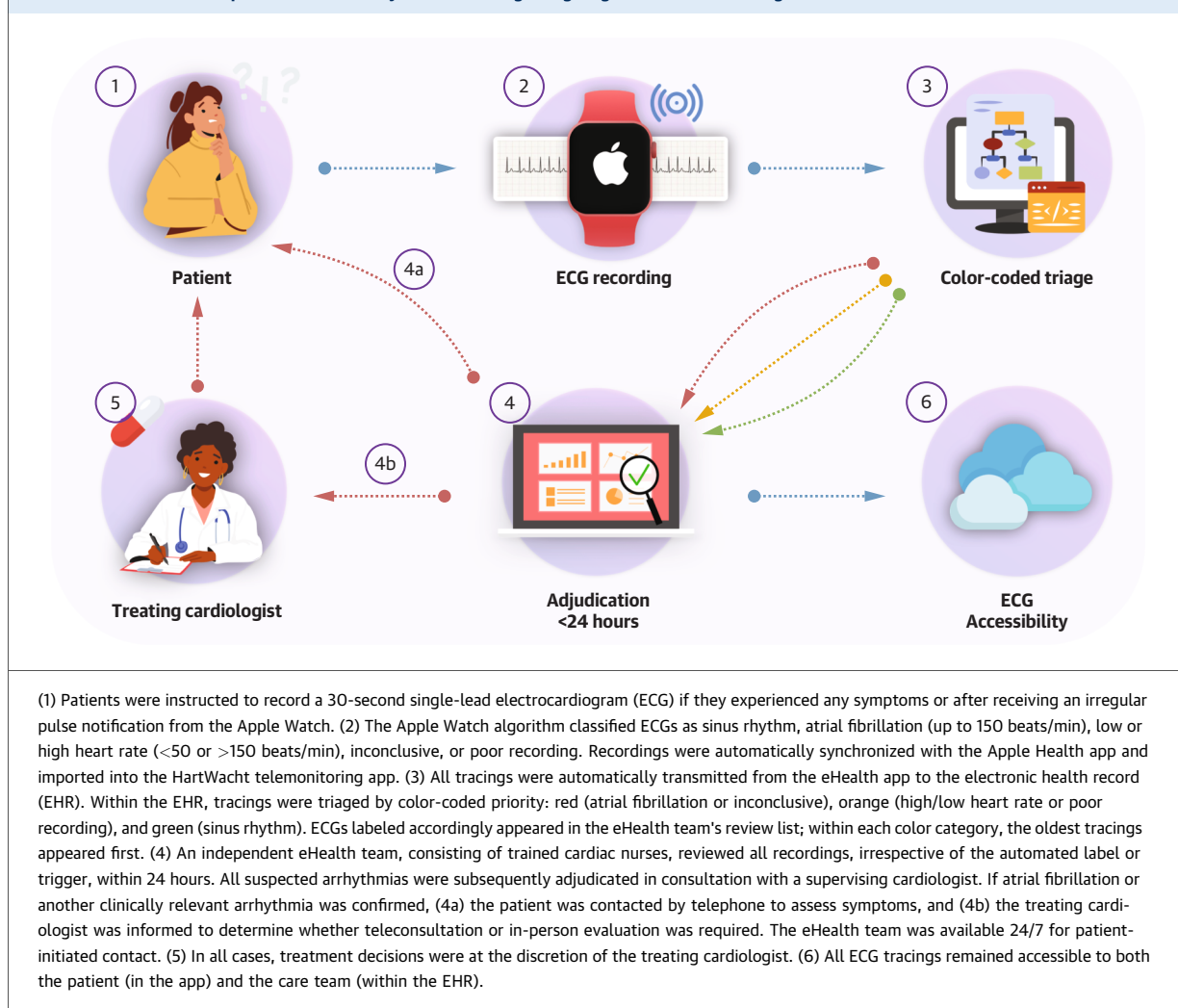
Baseline characteristics were collected from the electronic health record and included age, sex, medical history, including CHA₂DS₂-VASc score components, and current medication use.

During follow-up, data were collected on smart-watch adherence, ECG submissions, and AF detection and management. Adherence was verified by cross-checking ECG transmissions and device activity logs at the end of follow-up (see [Supplemental Methods](#) for details). Additionally, emergency department (ED) visits and major adverse cardiovascular events (MACE) were recorded.

OUTCOMES. The primary outcome was new onset AF within 180 days, defined as an AF episode lasting ≥ 30 seconds on single-lead ECG or 12-lead ECG (clinical AF). Secondary outcomes included the number of visits to the ED, the initiation of AF treatment, and MACE, including cardiovascular death, nonfatal myocardial infarction, and nonfatal stroke.

STATISTICAL ANALYSIS. Analyses were performed according to the intention-to-treat principle. Event

FIGURE 2 Schematic Representation of Study Procedures Regarding Single-Lead ECG Recording



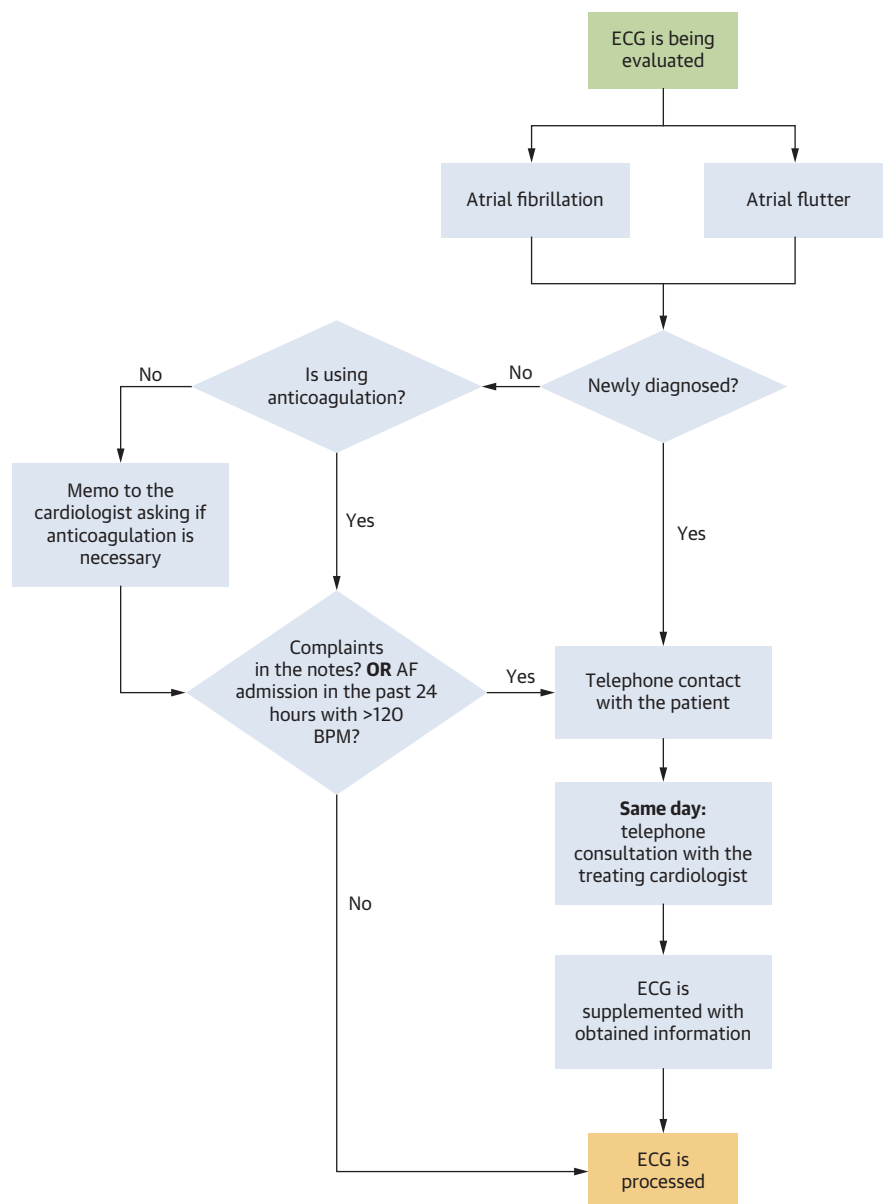
rates and 95% CIs were estimated from Kaplan-Meier curves. The between-group absolute risk difference at 180 days (intervention minus control) was calculated using the method of Com-Nougue et al,²² with the *z* statistic estimated from Kaplan-Meier event rates and Greenwood's standard errors; 2-sided *P* values were derived from the corresponding Wald test, and the difference is reported in percentage points. Cox proportional hazards regression, with trial center as a covariate, was used to estimate HRs and 95% CIs. The proportional hazards assumption was evaluated using Schoenfeld residuals; no significant violations were observed. Follow-up was censored at the time of death or at 180 days in patients who did not experience the primary endpoint.

ED visits were analyzed using a 2 × 2 contingency table comparing the proportion of patients with ≥1 ED visit between groups; effect estimates

are reported as ORs with 95% CIs. Initiation of AF treatment and MACE were summarized descriptively. Descriptive statistics are presented as mean ± SD or median (Q1-Q3) for continuous variables and as counts and percentages for categorical variables. No missing data were observed for any of the key variables in the trial. A 2-tailed probability value of <0.05 was considered statistically significant.

Continuous monitoring with an ILR detects new onset AF in 20.4% of high-risk patients within 6 months.⁹ Assuming that the AW, worn 12 hours/day, would detect 50% of these cases, the expected detection rate in the intervention group was 10.2%. Based on prior studies reporting 0.5% to 1.5% AF detection in control groups, the expected rate in the control arm was set at 1.2%.^{11,13} The trial was powered for superiority on the between-group absolute

FIGURE 3 HartWacht Telemonitoring Workflow for AF



Flowchart illustrating the clinical review pathway within the HartWacht telemonitoring platform for single-lead electrocardiograms (ECGs) classified as atrial fibrillation (AF) or atrial flutter by the eHealth team. Source: HartWacht protocol. BPM = beats/min.

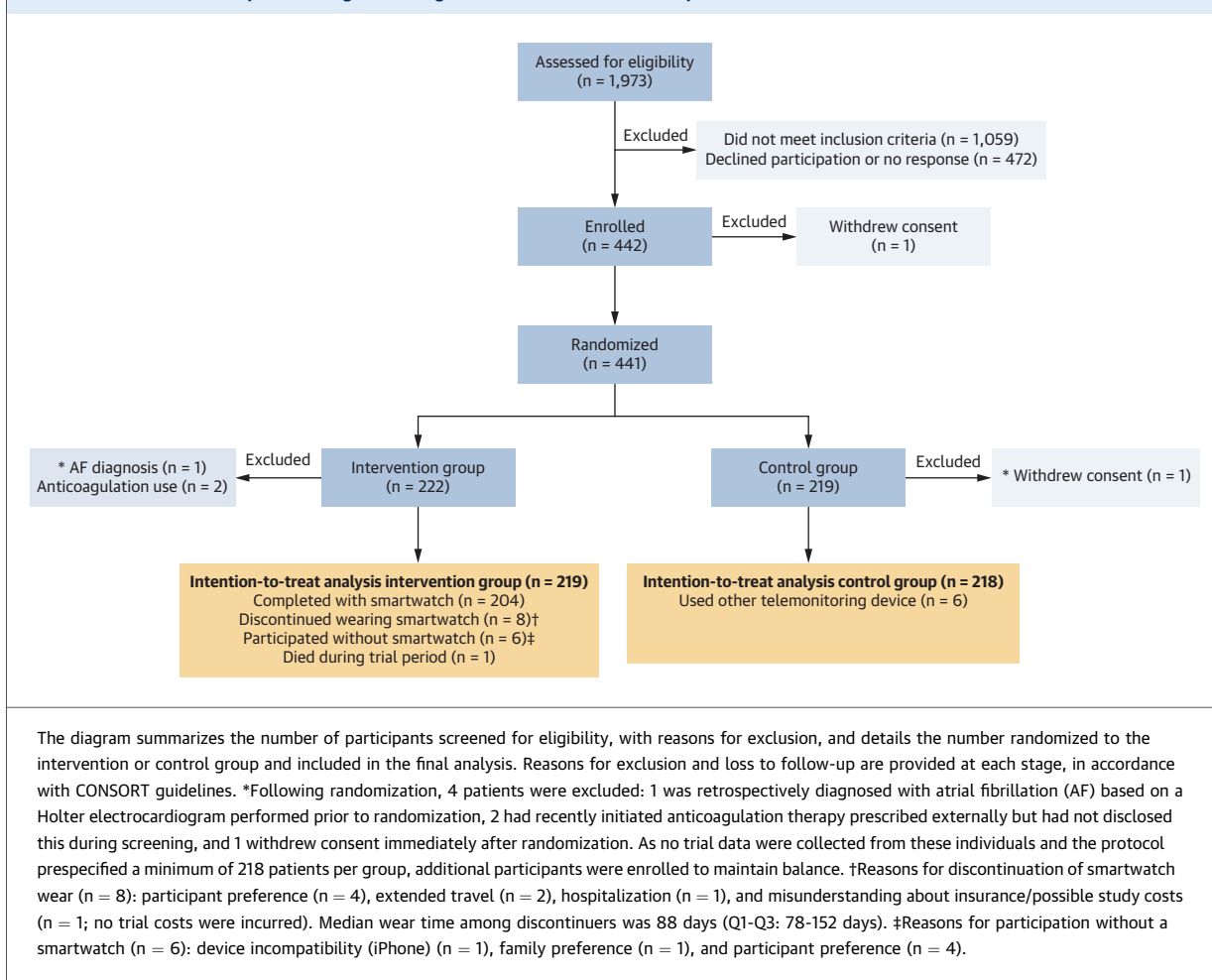
risk difference in new onset AF at 180 days. To detect a 9-percentage-point difference with 90% power at a 2-sided α of 0.05, 436 participants (218 per group) were required.

All statistical analyses were conducted using IBM SPSS Statistics, version 28.0.1.1. Kaplan-Meier survival curves were generated in Python version 3.11.5

(Python Software Foundation) using the Lifelines package version 0.30.0.

The trial was reported in accordance with the CONSORT guidelines for randomized controlled trials. Three scheduled monitor visits were conducted during the trial to ensure adherence to the trial protocol, verify data

FIGURE 4 Flow of Participants Through Screening, Randomization, and Follow-Up



accuracy, and confirm compliance with regulatory requirements.

RESULTS

STUDY POPULATION AND BASELINE CHARACTERISTICS.

Between November 2022 and December 2024, a total of 1,973 patients were assessed for eligibility. Of these patients, 1,059 did not meet inclusion criteria and 472 declined participation.

A total of 442 patients were enrolled. One patient withdrew informed consent prior to randomization, leaving 441 patients eligible for randomization. Four patients were excluded after randomization due to protocol violations or immediate withdrawal of consent. As no trial data were collected from these patients and the protocol specified 218 patients per group to ensure adequate power, additional

participants were enrolled to maintain balance between both groups. The final randomized population included 437 patients (219 intervention, 218 control). A detailed overview of patient inclusion and exclusion is provided in [Figure 4](#).

The median age of the trial population was 75 years (Q1-Q3: 71.0-78.0 years), and 204 (46.7%) patients were female. The median CHA₂DS₂-VASc score was 3.0 (Q1-Q3: 3.0-4.0). Baseline characteristics and reasons for initial referral to the cardiology outpatient clinic are presented in [Tables 1 and 2](#), respectively; standardized mean differences for baseline variables are shown in [Supplemental Table 1](#).

In the intervention group, 6 patients did not receive a smartwatch ([Figure 4](#)), due to smartphone incompatibility, anxiety about being monitored, or family concerns about electronic devices. Eight patients discontinued smartwatch use prematurely

but completed follow-up without the device (median wear time 88 days [Q1-Q3: 78-152 days]). Baseline characteristics of patients with protocol deviations are summarized in [Supplemental Table 2](#). One patient died during follow-up due to an unrelated cause.

Patients submitted a median of 21 ECGs (Q1-Q3: 10-42 ECGs) during the 6-month follow-up (n = 213). Those who discontinued wearing the smartwatch prematurely submitted a median of 17 ECGs (Q1-Q3: 9-47 ECGs) (n = 8).

In total, 60 (27.4%) patients in the intervention group and 50 (22.9%) in the control group underwent standard forms of rhythm monitoring during follow-up ([Table 3](#)). Patient-initiated use in the control group included 1 patient who used an AW with ECG functionality, though the irregular rhythm detection feature was disabled, and 4 (1.8%) who used smart devices from other vendors. No patient-initiated devices were connected to remote monitoring programs. **PRIMARY OUTCOME.** The primary endpoint, new onset AF within 180 days, occurred in 21 patients in the intervention group and in 5 patients in the control group. At 180 days, the Kaplan-Meier cumulative incidence of new onset AF was 9.6% in the intervention group vs 2.3% in the control group, yielding an absolute risk difference of 7.3 percentage points (95% CI: 2.9-11.7 percentage points; $P = 0.001$). Time-to-event analysis estimated an HR of 4.40 (95% CI: 1.66-11.66), indicating earlier AF detection in the intervention group. Kaplan-Meier curves are shown in [Figure 5](#).

Results of exploratory sensitivity analyses (per protocol, as treated, and excluding poststroke referrals), and adjusted for age, sex, hypertension, and trial center yielded results consistent with the primary intention-to-treat findings ([Supplemental Table 3](#)). Event rates translated to a number needed to screen of 14 (95% CI: 9-34), and when normalized to observation time, to approximately 1.6 new AF diagnoses per 100 patient-months.

In the intervention group, all initial AF diagnoses were based on single-lead ECG recordings. Those who were diagnosed with AF submitted a median of 6 ECGs (Q1-Q3: 3-11 ECGs) before AF was first detected. [Figure 6](#) shows the number of smartwatch ECGs submitted by each patient, stratified by AF diagnosis. One tracing was labeled as tachycardia (157 beats/min) by the AW algorithm but was interpreted as AF by the eHealth team. One patient who was not diagnosed with AF missed 2 irregular

TABLE 1 Demographic and Clinical Characteristics of the Patients at Baseline

	Intervention Group (n = 219)	Control Group (n = 218)
Age, y	75.0 (71.0-78.0)	75.0 (71.0-78.0)
Female	90 (41.1)	114 (52.3)
BMI, kg/m ²	26.2 (23.8-28.7)	25.7 (23.7-28.3)
CHA ₂ DS ₂ -VASc score ^a		
Median (Q1-Q3)	3.0 (3.0-4.0)	3.0 (3.0-4.0)
Score ≥4	103 (47.0)	104 (47.7)
History of hypertension	142 (64.8)	162 (74.3)
Diabetes mellitus	50 (22.8)	29 (13.3)
History of heart failure	11 (5.0)	12 (5.5)
History of vascular disease ^b	79 (36.1)	65 (29.8)
History of systemic embolism, stroke, or TIA	44 (20.1)	40 (18.3)
Comorbidities		
Chronic pulmonary disease	16 (7.3)	19 (8.7)
Obstructive sleep apnea	16 (7.3)	15 (6.9)
Dyslipidemia	83 (38.1)	80 (36.7)
Thyroid disorders	14 (6.5)	20 (9.2)
Myocardial infarction	37 (17.0)	32 (14.7)
Baseline antiarrhythmic medication use		
Beta-blocking agents	77 (35.2)	71 (32.6)
Calcium-channel blocking agents	10 (4.6)	15 (6.9)
Sodium-channel blocking agents	1 (0.5)	1 (0.5)

Values are median (Q1-Q3) or n (%). ^aCHA₂DS₂-VASc scores range from 0 to 9, with higher scores indicating a higher risk of stroke among patients with atrial fibrillation. ^bAngiographically significant coronary artery disease, previous myocardial infarction, peripheral artery disease, or complex aortic plaque.
BMI = body mass index; TIA = transient ischemic attack.

rhythm notifications while asleep and did not record an ECG in response.

Of 37 patients with at least 1 AW ECG labeled as AF, 20 patients were clinically diagnosed with AF, confirmed by the eHealth adjudication team and the treating physician based on ECG review, while 17 were not, resulting in a positive predictive value (PPV) of 54%. [Figures 7 and 8](#) illustrate AW algorithm classifications and corresponding clinical adjudication outcomes (confirmed AF vs no AF).

In the control group, AF was diagnosed in 1 patient using Holter monitoring, in 2 using telemetry observation during a hospital admission and ED visit, and in 2 using a different handheld telemonitoring device. Among patients diagnosed with AF, all patients in the control group experienced symptoms during their AF episode, whereas 9 (42.9%) of 21 patients in the intervention group reported symptoms.

INITIATION OF AF TREATMENT. All patients diagnosed with AF received anticoagulant therapy. An

TABLE 2 Reason for First Referral to Cardiology Center

Reason for Referral	Intervention Group (n = 219)	Control Group (n = 218)
Suspected or known ischemic heart disease		
Angina pectoris	83 (37.9)	80 (36.7)
Checkup post-PCI or post-CABG	14 (6.4)	12 (5.5)
Structural or valvular heart disease		
Structural heart disease or murmur evaluation	7 (3.2)	15 (6.9)
Suspected heart failure or miscellaneous		
Dyspnea (on exertion)	20 (9.1)	22 (10.1)
Fatigue	5 (2.3)	5 (2.3)
Reduced exercise capacity	2 (0.9)	5 (2.3)
Suspected heart failure	3 (1.4)	2 (0.9)
Arrhythmic symptoms		
Palpitations or tachycardia ^a	21 (9.6)	21 (9.6)
Bradycardia	6 (2.7)	1 (0.5)
Premature beats (symptomatic/asymptomatic)	5 (2.3)	3 (1.4)
Syncope or presyncope		
Presyncope/vasovagal/orthostatic complaints (without TLOC)	9 (4.1)	5 (2.3)
Reflex syncope (with TLOC, vasovagal/orthostatic)	3 (1.4)	2 (0.9)
Syncope with features indicating higher arrhythmic risk ^a	2 (0.9)	1 (0.5)
Evaluation of cardioembolic source ^a		
Evaluation post-CVA	2 (0.9)	3 (1.4)
Evaluation post-TIA	14 (6.4)	7 (3.2)
Cardiovascular risk management and screening		
Hypertension	11 (5.0)	15 (6.9)
Prevention or screening	12 (5.5)	17 (7.8)
Other	0 (0.0)	2 (0.9)

Values are n (%). Reason for referral reflects the initial historical referral to a secondary care cardiology center and may predate randomization by years. ^aAmong patients referred for palpitations or tachycardia, 20 (95.2%) of 21 in the intervention group and 16 (76.2%) of 21 in the control group underwent Holter monitoring after their initial referral. In addition, all patients referred for evaluation of a cardioembolic source after CVA or TIA, and those referred for syncope with features indicating higher arrhythmic risk also underwent Holter monitoring before trial enrollment.

CABG = coronary artery bypass grafting; CVA = cerebrovascular accident; PCI = percutaneous coronary intervention; TIA = transient ischemic attack; TLOC = transient loss of consciousness.

TABLE 3 Follow-Up Physician-Prescribed Monitoring Exposure

Monitoring Type	Intervention Group (n = 219)	Control Group (n = 218)
Any follow-up monitoring ^a	60 (27.4)	50 (22.9)
12-lead ECG only	47 (21.5)	25 (11.5)
Holter monitoring only	5 (2.3)	12 (5.5)
Both 12-lead ECG and Holter monitoring	3 (1.4)	5 (2.3)
In-hospital telemetry only	1 (0.5)	0 (0.0)
Both in-hospital telemetry and 12-lead ECG	3 (1.4)	2 (0.9)
Both in-hospital telemetry and Holter monitoring	1 (0.5)	0 (0.0)
Handheld telemonitoring device only	0 (0.0)	4 (1.8)
Both handheld telemonitoring device and 12-lead ECG	0 (0.0)	2 (0.9)

Values are n (%). Categories are mutually exclusive. ^aIncludes all physician-prescribed modalities: 12-lead ECG, Holter monitoring, in-hospital telemetry, and handheld telemonitoring.

ECG = electrocardiogram.

overview of AF-related symptoms, subtype classification (paroxysmal, persistent, permanent AF), and initiation of AF treatment is provided in [Table 4](#). Detailed, individual-level information on each AF case is available in [Supplemental Tables 4A and 4B](#).

ED VISITS AND MACE. During the follow-up period, 13 (5.9%) patients in the intervention group and 18 (8.3%) patients in the control group visited the ED (OR: 0.70; 95% CI: 0.34-1.47) ([Supplemental Table 5](#)). Four patients (2 in each group) visited the ED twice, both times for the same reason. MACE occurred in 2 (0.9%) patients in each group: 1 stroke and 1 cardiovascular death in the intervention group and 2 non-ST-segment elevation myocardial infarctions in the control group.

DISCUSSION

In this randomized controlled trial, 6-month smartwatch-based rhythm monitoring significantly increased the detection of new onset AF (9.6% vs 2.3%; absolute risk difference: 7.3 percentage points; 95% CI: 2.9-11.7 percentage points; HR: 4.40; 95% CI: 1.66-11.66) and shortened the time to diagnosis compared with standard care. To our knowledge, this is the first randomized controlled trial of smartwatch-based AF screening in a high-stroke-risk population.²³ The European Society of Cardiology guidelines suggest tools for AF screening, using handheld and smart devices among others, but no clear prolonged screening routine is recommended. Our findings demonstrate that prolonged screening is feasible, using a workflow-integrated smartwatch-based screening method, combining PPG and single-lead ECG functions, to identify undiagnosed AF in patients at highest clinical risk.

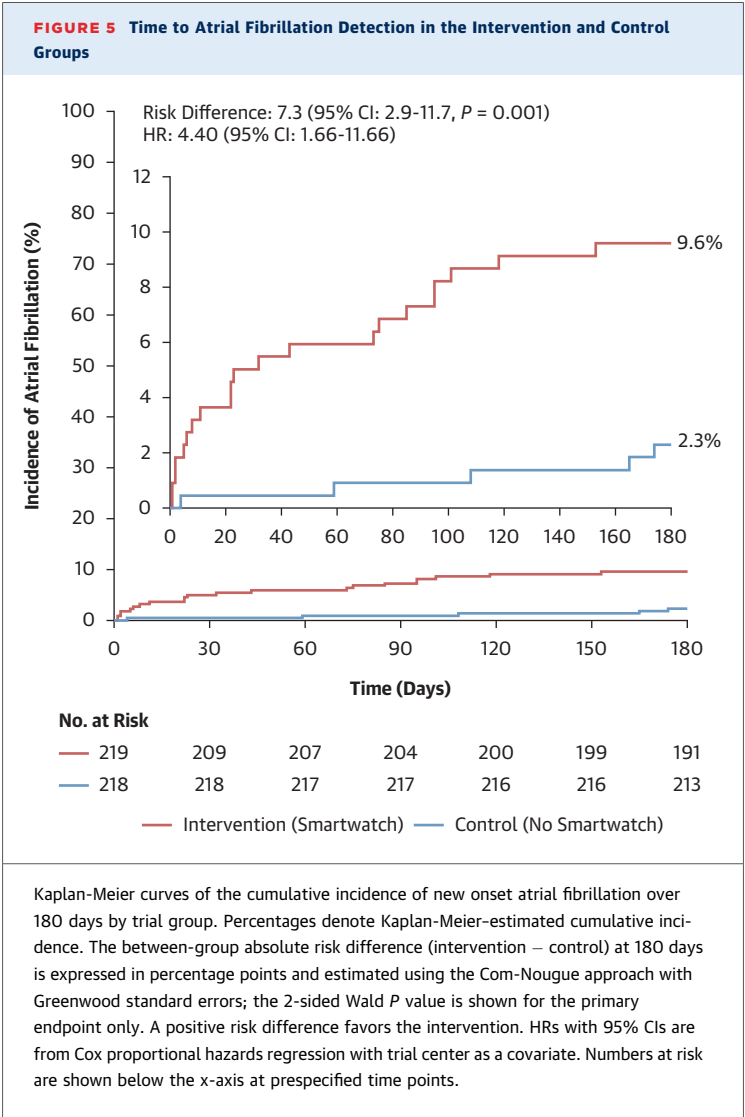
The level of evidence supporting routine or population-based AF screening remains low (European Society of Cardiology guidelines AF 2024). The STROKESTOP (Systematic ECG Screening for Atrial Fibrillation Among the Elderly) trial demonstrated a positive effect of prolonged noninvasive screening on clinical outcomes, whereas prolonged invasive AF screening in the Atrial fibrillation detected by continuous electrocardiographic monitoring using implantable loop recorder to prevent stroke in individuals at risk (LOOP) trial showed no such benefit.^{12,24} Several outcome-driven trials, including the REGAL (Realtime Diagnosis from Electrocardiogram Artificial Intelligence-Guided Screening for Atrial Fibrillation with Long Follow-Up) and SAFER (Screening for Atrial Fibrillation with ECG to Reduce Stroke) trials, are ongoing to determine whether early detection and treatment of screen-detected AF

reduce the risk of stroke, heart failure admissions, or cognitive decline.^{25,26}

Patients with implantable devices were excluded from the current trial, as this distinctly different population often has underlying structural heart disease and pacing indications. Previous device-based trials, including the ARTRESIA (Apixaban for the Reduction of Thrombo-Embolism in Patients With Device-Detected Subclinical Atrial Fibrillation) trial and the NOAH-AFNET 6 (Non-vitamin K Antagonist Oral Anticoagulants in Patients With Atrial High Rate Episodes) trial, demonstrated only modest reductions in stroke risk, accompanied by an increased, mostly reversible, bleeding risk.^{27,28}

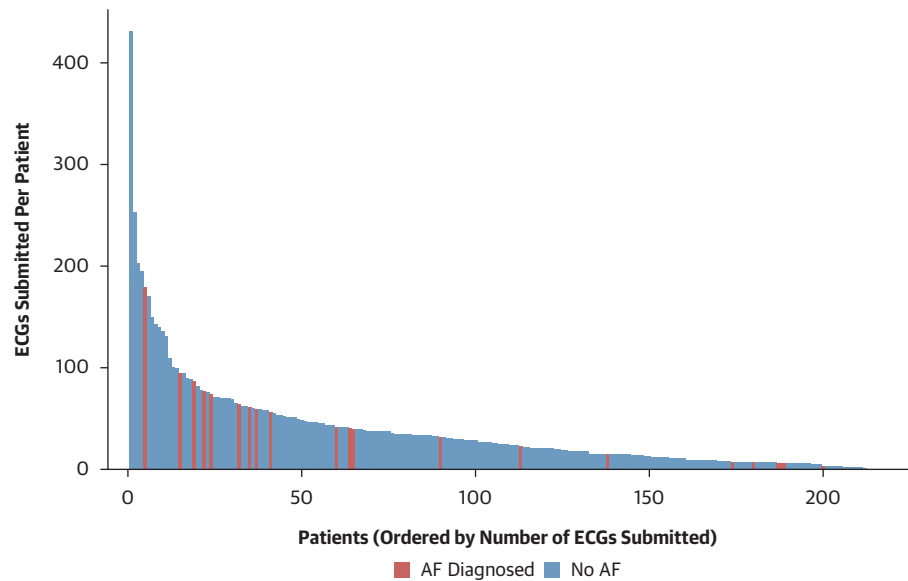
Although the current trial was not designed to assess outcome benefits, it demonstrates the feasibility of prolonged AF screening in nondevice patients at elevated stroke risk in a real-world setting. Prior AF screening studies have evaluated a wide range of monitoring strategies, each with distinct trade-offs between diagnostic yield, invasiveness, and patient burden. Continuous long-term monitoring with ILRs provides the highest diagnostic yield, with approximately a 3-fold increase in AF detection compared with standard care, but their invasive nature restricts use to highly selected patients.²⁴ The SCREEN-AF (Screening for Atrial Fibrillation in the Older Population) trial showed that a 2-week continuous ECG patch increased AF detection 10-fold (5.3% vs 0.5%), although adherence was limited by adverse skin reactions.¹³ In contrast, the REHEARSE-AF (Assessment of Remote Heart Rhythm Sampling Using the AliveCor Heart Monitor to Screen for Atrial Fibrillation) trial, which used handheld single-lead ECG devices for twice-weekly and symptom-triggered recordings, demonstrated a 4-fold higher yield (3.8% vs 1.0%) over 12 months but relied on regular user activation and was therefore less effective for asymptomatic or nocturnal AF.¹¹ Similarly, the electronic eBRAVE-AF (BRoad-based Atrial fibrillation screening using a VERification algorithm) trial, using a smartphone PPG app, approximately doubled AF detection (~1% over 6 months) yet depended on patient-initiated measurements and delayed patch ECG monitoring for confirmation.

In contrast, the EQUAL trial integrated smartwatch-based AF detection within a centralized 24/7 telemonitoring infrastructure, enabling automatic ECG transmission, professional adjudication, and prompt clinical feedback. This coordinated workflow, combining passive PPG-based rhythm



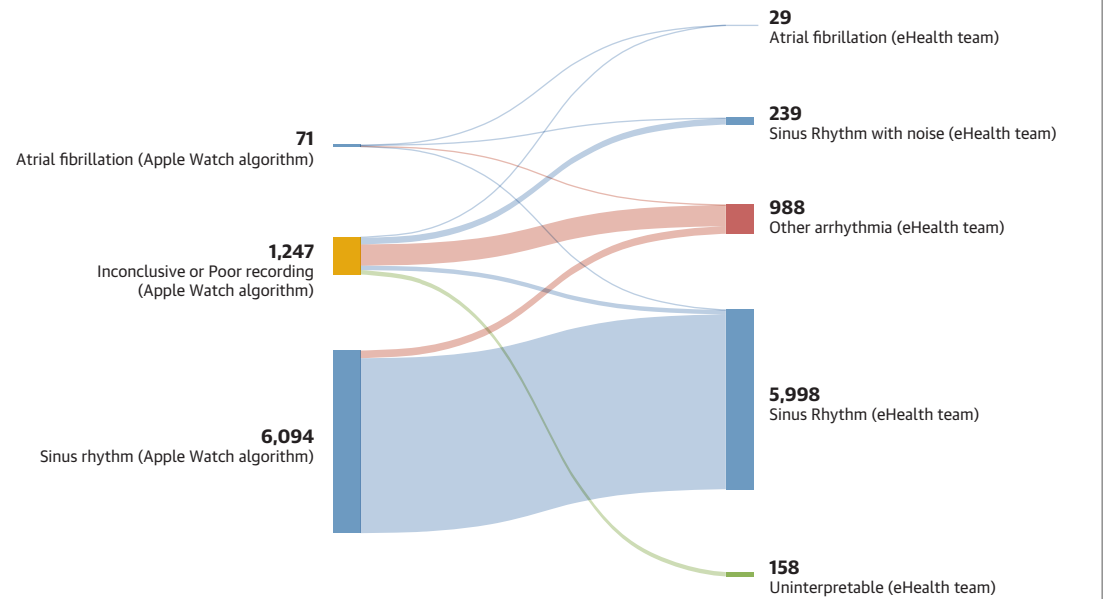
surveillance with on-demand ECG confirmation, provided a feasible and scalable model for prolonged AF screening in a real-world high-risk population, capable of identifying both symptomatic and asymptomatic cases. Moreover, the absence of adverse skin reactions from smartwatch use supported sustained participant engagement and enabled prolonged observation. This approach, when normalized to observation time, yielded an approximate 1.6 new AF diagnoses per 100 patient-months, vs 0.32 in the REHEARSE-AF trial and 0.22 in the eBRAVE-AF trial, suggesting that prolonged, partly passive monitoring can achieve a higher yield within a shorter time frame. Notably, the incidence of new onset AF in both trial arms was higher than in

FIGURE 6 Distribution of Smartwatch ECG Submissions per Patient

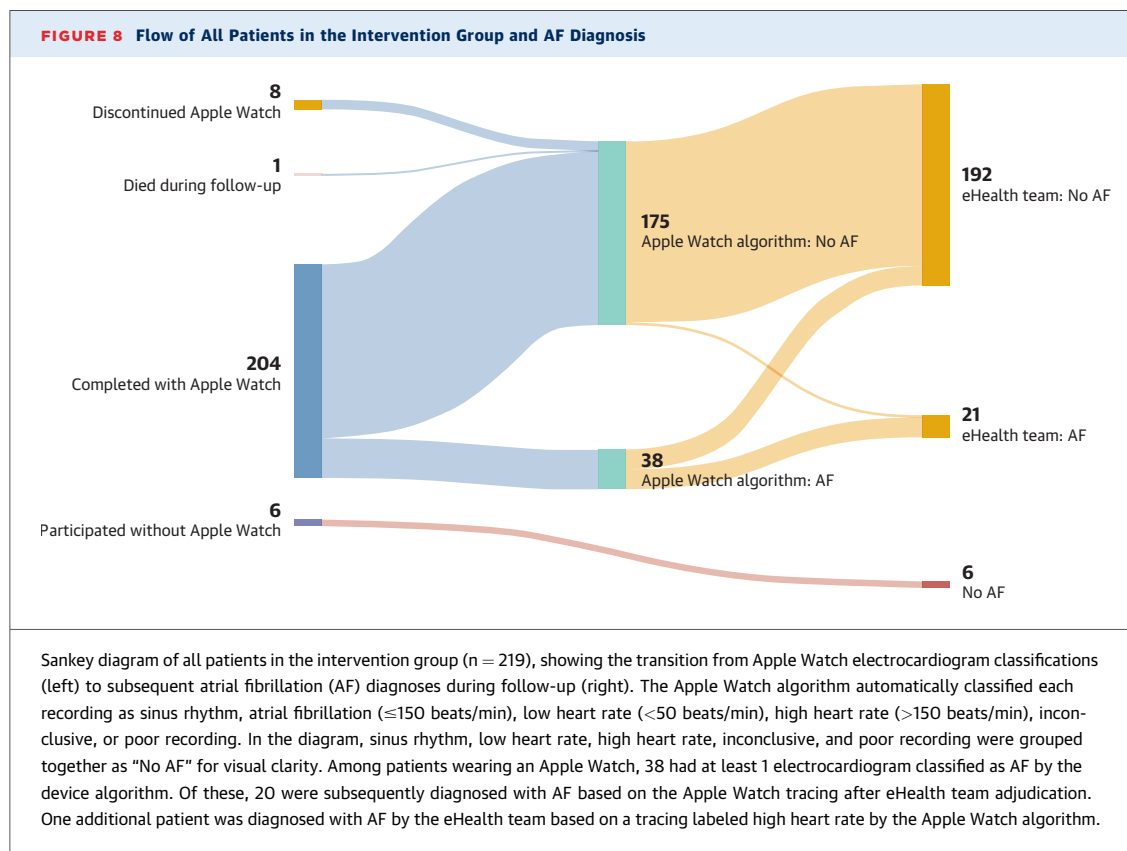


Each bar represents 1 patient ($n = 213$), ordered by the total number of smartwatch ECGs submitted during follow-up. Patients diagnosed with AF ($n = 21$) are highlighted in red and those without AF are highlighted in blue. Bars represent all ECGs transmitted to the telemonitoring platform before censoring at 6 months or at AF diagnosis. Abbreviations as in [Figure 3](#).

FIGURE 7 Flow of Apple Watch Electrocardiogram Classifications to Clinical Adjudication Outcomes



Sankey diagram depicting the transition from automated Apple Watch electrocardiogram classifications (left) to final clinical adjudications (right). The Apple Watch algorithm automatically classified each recording as sinus rhythm, atrial fibrillation (≤ 150 beats/min), low heart rate (< 50 beats/min), high heart rate (> 150 beats/min), inconclusive, or poor recording. In the diagram, low heart rate, high heart rate, inconclusive, and poor recording classifications were grouped together under “Inconclusive or Poor recording,” as these outputs do not provide a rhythm diagnosis (ie, they do not distinguish between sinus rhythm and atrial fibrillation). According to the trial protocol, the eHealth team classified each tracing as sinus rhythm (including sinus rhythm with noise when baseline artefacts did not preclude interpretation), atrial fibrillation, other arrhythmias, or uninterpretable when assessment was not possible. The width of each stream corresponds to the number of tracings within that classification flow.



previous trials, which may partly be attributable to differences in the trial population. While prior studies primarily enrolled participants from primary care or community settings, our trial included secondary-care cardiology patients, who likely had a higher baseline risk of AF. Nonetheless, inclusion criteria and baseline characteristics, including median age and CHA₂DS₂-VASc score, were broadly comparable across studies.

In addition to the higher detection yield, the nature of AF detected provides further insight into the performance of smartwatch-based screening. In the REHEARSE-AF trial, 12 (63%) of 19 patients had paroxysmal AF and 8 (42%) were asymptomatic after 12 months of handheld ECG monitoring. In our intervention group, 19 (91%) of 21 AF cases were paroxysmal and 12 (57%) were asymptomatic, compared with 4 (80%) and 0 (0%) cases in the control arm, respectively. This higher proportion and absolute number of silent, paroxysmal cases likely reflect the ability of smartwatch monitoring to capture brief, self-terminating episodes that symptom-based approaches may miss. Importantly, accumulating evidence indicates that asymptomatic AF carries a thromboembolic and mortality risk

TABLE 4 Patients With AF

AF Characteristic	Intervention Group (n = 219)	Control Group (n = 218)
Any AF diagnosis	21 (9.6)	5 (2.3)
Paroxysmal AF	19 (90.5)	4 (80.0)
Persistent AF	2 (9.5)	1 (20.0)
Permanent AF	0 (0.0)	0 (0.0)
Symptoms status at time of AF diagnosis		
Asymptomatic	12 (57.1)	0 (0.0)
Symptomatic	9 (42.9)	5 (100.0)
Palpitations ($\pm$ dyspnea, fatigue, or dizziness)	6 (28.6)	4 (80.0)
Fatigue only	2 (9.5)	0 (0.0)
Dyspnea only	1 (4.8)	0 (0.0)
Left-sided heart failure	0 (0.0)	1 (20.0)
Management		
Anticoagulation	21 (100.0)	5 (100.0)
Antiarrhythmic or beta-blocker	6 (28.6)	3 (60.0)
Electrocardioversion	3 (14.3)	0 (0.0)
Ablation	1 (4.8)	0 (0.0)

Values are n (%). AF subtype based on European Society of Cardiology criteria (2020): paroxysmal AF was defined as AF episodes that terminated spontaneously or with intervention within 7 days of onset; persistent AF lasted > 7 days or required pharmacological or electrical cardioversion after that period; permanent AF was diagnosed when a joint decision was made not to pursue rhythm control. Symptomatic AF was defined as presence of palpitations, dizziness, syncope, or chest discomfort at the time of AF detection.

AF = atrial fibrillation.

comparable to symptomatic AF, reinforcing the need for early identification and guideline-directed management once AF is confirmed.²⁹

Full workflow integration is essential to scale screening endeavors, without further straining our health care systems. Our trial uniquely demonstrates how a consumer-grade wearable can be embedded into a clinical telemonitoring infrastructure to enable rapid adjudication and safe implementation in real-world care. In the current setup all smartwatch ECGs, regardless of automated classification by the smartwatch algorithm, triggered review by a dedicated eHealth team, ensuring that AF was confirmed before clinical action.

The smartwatch algorithm's PPV of approximately 54% is consistent with expectations for front-end automated classifications and comparable to rates observed in standard ECG workflows, in which computer-interpreted "AF" diagnoses are incorrect in about 9% of cases. The lower PPV compared with prior digital screening studies using PPG with delayed ECG patch confirmation likely reflects differences in confirmatory strategies, stricter adjudication criteria (ie, requiring AF to be present throughout a full 30-second ECG), and differences in population characteristics, including a higher prevalence of ectopic beats or borderline rhythms in this high-risk cardiology cohort. Additionally, smartwatch ECGs were recorded in an unsupervised setting and are more prone to motion artifacts and transient noise, which may further reduce automated accuracy. By incorporating clinical adjudication into the detection cascade, our trial minimized the risk of inappropriate anticoagulation or overtreatment due to false positive classifications. All confirmed AF cases led to guideline-directed management, including oral anticoagulation in 100 % of patients and, when appropriate, rhythm-control strategies. These findings underscore that workflow integration should pair device algorithms with clinician oversight and structured care pathways.

It is important, however, to further prioritize equitable access, for example by providing temporary loan devices with structured support through existing telemonitoring programs. This would allow general access to AF screening, and facilitate inclusion of diverse patient populations.

STUDY LIMITATIONS. The open-label design without a sham device may have introduced behavioral bias, though objective endpoints and independent adjudication minimized this risk. The number of AF events

was modest, limiting statistical precision. As follow-up ended at the time of AF diagnosis and only discrete 30-second ECG recordings were available, exact episode duration and recurrence could not be assessed. However, AF was diagnosed on single-lead ECG only when a complete 30-second recording showed AF. Because the trigger for ECG recordings was not visible to the telemonitoring team, we were unable to determine how many were submitted in response to symptoms vs irregular-rhythm notifications. This limitation did not affect clinical classification of AF, as classification was based solely on adjudicated ECG findings and the presence or absence of symptoms, independent of the trigger prompting ECG submission. Our trial was not powered for long-term clinical outcomes, and further research is needed to establish how AF burden and downstream events should guide management decisions in screen-detected AF. Finally, patients were recruited from specialty cardiology clinics and all possessed, or loaned, compatible smartphones, which may limit generalizability to more diverse populations.

In summary, this trial demonstrates that smartwatch-based AF screening in high-risk patients is feasible, safe, and markedly increases AF detection compared with standard care. It illustrates how consumer wearables can be embedded into telemonitoring workflows to deliver rapid, verified diagnosis and timely initiation of therapy. While these results establish feasibility and improved detection, the clinical benefit of treating screen-detected AF awaits confirmation in ongoing outcome trials. Together, these findings position smartwatch-based screening as a practical bridge between population-based digital detection and evidence-based treatment.

CONCLUSIONS

This randomized controlled trial provides strong evidence that smartwatch-based AF screening using PPG and ECG functions enhances the detection rate of new onset AF compared with standard care in patients at elevated stroke risk. By embedding consumer wearables into telemonitoring workflows, this trial provides a scalable model for integrating digital screening into routine cardiology practice. While this trial establishes feasibility and diagnostic benefit, the clinical impact of treating screen-detected or subclinical AF awaits confirmation in ongoing outcome trials.

DATA AVAILABILITY STATEMENT The data that support the findings of this trial and the trial protocol

are available from the corresponding authors upon reasonable request.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: This randomized trial demonstrates that smartwatch-based screening using both PPG and ECG functions can identify undiagnosed, often asymptomatic and paroxysmal AF in individuals at elevated stroke risk, providing a feasible, real-world approach to noninvasive rhythm monitoring.

COMPETENCY IN PATIENT CARE AND PROCEDURAL SKILLS: Earlier detection of AF enables timely, guideline-directed initiation of anticoagulation and rhythm-control strategies in patients most likely to benefit, though the outcome benefit of treating screen-detected AF remains to be proven.

COMPETENCY IN SYSTEMS-BASED PRACTICE: Integration of smartwatch-based AF screening in routine care requires structured telemonitoring framework, clinician oversight, and strategies to address disparities in digital literacy and device accessibility, ensuring equitable implementation, particularly in older populations.

TRANSLATIONAL OUTLOOK 1: Future studies should define the optimal screening duration and monitoring intensity to maximize diagnostic yield and cost-effectiveness.

TRANSLATIONAL OUTLOOK 2: Outcome-driven trials such as REGAL and SAFER will determine whether earlier detection and treatment of screen-detected AF can translate into reductions in stroke, heart failure, or cognitive decline.

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KEY WORDS atrial fibrillation, digital health, randomized controlled trial, remote monitoring, wearable technology

APPENDIX For an expanded Methods section and supplemental tables, please see the online version of this paper.